南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目：	概率论与数理统计	**开课单位：**	数学系
考试时长：	2025/4/19 19:00–21:00	**命题教师：**	概率统计教学组

本试卷共三大部分，满分（100）分
（考试结束后请将试卷、答题卡一起交给监考老师）

题号	I	II	III–1	III–2	III–3	III–4	III–5	III–6
分值	20 分	20 分	10 分	10 分	10 分	10 分	10 分	10 分

第I部分 选择题（每题4分，总共20分）

Part I – Single Choice (4 marks each question, 20 marks in total)

1. 将7个人分成3组来执行相同的任务, 其中一组有3个人, 另外两组各有2个人. 有多少种分组方式?

   Divide 7 people into 3 groups to perform the same task, with one group of 3 and the other two groups of 2 each. How many ways are there to form the groups?

   A. 35 **B.** 210 **C.** 105 **D.** 70

2. 随机事件A 和B 满足$P(A)=P(B)=\frac{1}{2}$且$P(A\cup B)=1$, 那么以下哪一项是正确的?

   Suppose random events A and B satisfy $P(A)=P(B)=\frac{1}{2}$ and $P(A\cup B)=1$, then which of the following is true?

   A. $A\cup B=\Omega$ **B.** $P(A-B)=0$ **C.** $P(\overline{A}\cup\overline{B})=1$ **D.** $A\cap B=\emptyset$

3. 某种医学检测会给出阳性或阴性结果. 阳性结果旨在表明一个人患有某种（罕见）疾病, 而阴性结果旨在表明他们没有该疾病. 然而, 假设该检测有时会给出错误的结果: 在没有该疾病的人中, 有1/100 的人会得到阳性检测结果; 而在患有该疾病的人中, 有2/100 的人会得到阴性检测结果. 如果每1000 人中有1 人患有该疾病, 那么一个检测结果为阳性的人实际上患有该疾病的概率是?

   A certain medical test either gives a positive or negative result. The positive test result is intended to indicate that a person has a particular (rare) disease, while a negative

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test result is intended to indicate that they do not have the disease. Suppose, however, that the test sometimes gives an incorrect result: 1 in 100 of those who do not have the disease have positive test results, and 2 in 100 of those having the disease have negative test results. If 1 person in 1000 has the disease, find the probability that a person with a positive test result actually has the disease?

A. $\frac{1097}{10000}$ **B.** $\frac{98}{1097}$ **C.** $\frac{1001}{10000}$ **D.** $\frac{98}{1007}$

4. 随机变量X的密度函数为$f_X(x)=\begin{cases}1-|x|, & -1\le x\le 1\\ 0, & \text{其他}\end{cases}$。求其分布函数在$x=\frac{1}{2}$处的值。

The probability density function of the random variable X is $f_X(x)=1-|x|$ for $-1\le x\le 1$. Find the value of its distribution function at $x=\frac{1}{2}$.

A. $\frac{1}{8}$ **B.** $\frac{3}{8}$ **C.** $\frac{5}{8}$ **D.** $\frac{7}{8}$

5. 给定联合密度函数$f(x,y)=\begin{cases}e^{-y}, & 0<x<y<\infty\\ 0, & \text{其他}\end{cases}$。以下哪一项是正确的？

Given the joint density function $f(x,y)=\begin{cases}e^{-y}, & 0<x<y<\infty\\ 0, & \text{otherwise}\end{cases}$. Which one of the following is true?

A. $f(3,2)=e^{-2}$ **B.** $f_X(x)f_Y(y)=f(x,y)$

C. $f_{Y|X}(y|x)=\begin{cases}e^{-(y-x)}, & y>x>0,\\ 0, & \text{otherwise.}\end{cases}$ **D.** $f_X(x)=\begin{cases}xe^{-x}, & x>0,\\ 0, & \text{otherwise.}\end{cases}$

第II部分 填空题（每空2分，总共20分）

Part II – Blank Filling (2 marks each blank, 20 marks in total)

1. 设A_1,A_2,A_3是样本空间的一个划分，且$P(A_1)=0.2$, $P(A_2)=0.5$, $P(B|A_1)=0.1$, $P(B|A_2)=0.4$, $P(B|A_3)=0.7$. 那么$P(A_3|B)=$_____.

Let A_1,A_2,A_3 be a partition of the sample space, with: $P(A_1)=0.2$, $P(A_2)=0.5$, $P(B|A_1)=0.1$, $P(B|A_2)=0.4$, and $P(B|A_3)=0.7$. Then $P(A_3|B)=$_____.

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2. 一个不透明的袋子中装有5个红球和3个白球, 每次取1球不放回，直到取出所有白球为止. 设停止时总取球次数为X，则$P(X=6)=$_____.
An opaque bag contains 5 red balls and 3 white balls. Balls are drawn one by one without replacement until all white balls are taken out. Let X be the total number of draws when the process stops. Then, the probability $P(X=6)=$_____.

3. 在长度为a的线段上, 于中点两侧各随机取一点, 将线段分成三段. 那么这三段长度不能构成三角形的概率是_____.
On a line segment of length a, two points are selected independently and uniformly at random, one on each side of the midpoint. This divides the segment into three parts. Then the probability that these three lengths cannot form a triangle is _____.

4. 设随机变量Y 服从参数为1 的指数分布，a 为常数且大于0，则$P\{Y \geq a+1 \mid Y > a\}=$_____.
Assume the random variable Y follows exponential distribution with the parameter 1. If a is a constant being greater than 0, then $P\{Y \geq a+1 \mid Y > a\}=$_____.

5. 假设离散型随机变量X的频率函数为$P(X=1)=0.6$, $P(X=0)=0.4$. 且当$X=1$时, $Y \sim U(0,2)$; 当$X=0$ 时, $Y \sim \text{Exp}(1)$. 那么$P(Y \leq 1)=$_____.
Assume that the frequency function of discrete random variable X is $P(X=1)=0.6$, $P(X=0)=0.4$. Additionally, we have that if $X=1$ then $Y \sim U(0,2)$; and if $X=0$ then $Y \sim \text{Exp}(1)$. Then $P(Y \leq 1)=$_____.

6. 设(X,Y)的联合密度为$f_{X,Y}(x,y)=6e^{-x}e^{-2y}$ $(0<x<y<+\infty)$. 则给定$X=x$时，Y的条件密度为$f_{Y|X}(y|x)=$_____$(y>x)$.
Let (X,Y) have the joint probability density function: $f_{X,Y}(x,y)=6e^{-x}e^{-2y}$ for $0<x<y<+\infty$. Then the conditional density is $f_{Y|X}(y|x)=$_____$(y>x)$.

7. 假设$X \sim N(\mu,4)$为一个正态分布随机变量. 已知$P(X \leq 1)=0.1587$, 则$\mu=$_____.(提示: $\Phi(1)=0.8413$, 其中Φ为标准正态分布函数)
Let $X \sim N(\mu,4)$ be a normally distributed random variable. Given $P(X \leq 1)=0.1587$, then $\mu=$_____. (*Hint:* $\Phi(1)=0.8413$, where Φ denotes the cumulative distribution function of $\mathcal{N}(0,1)$).

8. 设(X,Y)的联合密度函数为$f(x,y)=\begin{cases}k, & 0<x^2<y<x<1\\ 0, & \text{其他}\end{cases}$. 则常数$k=$_____.
Let (X,Y) be a two-dimensional random vector with joint probability density function given by $f(x,y)=\begin{cases}k, & \text{if } 0<x^2<y<x<1\\ 0, & \text{otherwise}\end{cases}$. Then the constant $k=$_____.

9. 设随机变量X与Y相互独立，X取值为1和2的概率均为0.5，Y取值为0和1的概率均为0.5. 令$Z=X+Y$，则$P(Z=2)=$_____.

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Let random variables X and Y be independent of each other. The probabilities that X takes the values of 1 and 2 are both 0.5, and the probabilities that Y takes the values of 0 and 1 are both 0.5. Let $Z = X + Y$, then $P(Z = 2) =$_____.

10. 设X 与Y 是两个随机变量，且$P\{X \geq 0, Y \geq 0\} = \frac{3}{7}$, $P\{X \geq 0\} = P\{Y \geq 0\} = \frac{4}{7}$. 则$P\{\max\{X, Y\} \geq 0\} =$_____.

Assume that X and Y are two random variables, and $P\{X \geq 0, Y \geq 0\} = \frac{3}{7}, P\{X \geq 0\} = P\{Y \geq 0\} = \frac{4}{7}$. Then $P\{\max\{X, Y\} \geq 0\} =$_____.

第III部分 解答题（每题10分，共60分）

Part III–Question Answering (10 marks each question, 60 marks in total)

1. 一个盒子中有4个红球、3个白球和2个黑球，编号分别为红球1-4、白球1-3、黑球1-2。现从中不放回地依次取出3个球，求：

(1) 第三次才取到红球的概率;

(2) 已知第二次取到白球的条件下，第一次取到黑球的概率;

(3) 三次取球中恰好有1个红球、1个白球、1个黑球（顺序不限）的概率.

A box contains: 4 red balls (numbered 1 - 4), 3 white balls (numbered 1 - 3), 2 black balls (numbered 1 - 2). Three balls are drawn without replacement in sequence. Find the following probabilities:

(1) The probability that the third ball is the first red ball drawn (i.e., the first two balls are not red, and the third is red).

(2) The conditional probability that the first ball is black, given that the second ball is white.

(3) The probability that the three balls drawn consist of exactly one red, one white, and one black ball (in any order).

2. 设E 和F 是某试验的样本空间中互不相容的事件，E 发生概率为$P1$，F 发生概率为$P2$。假定重复做该试验至E 和F 有一个发生。这个试验的样本空间是什么？事件E 在事件F 之前发生的概率是多少？

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Let E and F be mutually - exclusive events in the sample space of a certain experiment. The probability of E occurring is $P1$, and the probability of F occurring is $P2$. Suppose the experiment is repeated until either E or F occurs. What is the sample space of this experiment? What is the probability that event E occurs before event F ?

3. 某抽奖活动规则如下：

- 抽奖箱中有10张券，其中3张是一等奖，4张是二等奖，3张是谢谢参与。
- 玩家依次不放回地抽取2张券, X表示抽到一等奖的数量，Y表示抽到二等奖数量。

(1) 求(X,Y) 的联合概率分布表，并验证其合法性；

(2) 分别求X 和Y 的边缘分布，并判断独立性；

(3) 求抽到一二等奖数量之和的随机变量Z的概率分布。

A lottery event is conducted with the following rules:

- There are 10 tickets in a box: 3 are First Prize, 4 are Second Prize, and 3 are No Prize.
- A player draws 2 tickets without replacement. Let X be the number of First Prizes drawn, and Y the number of Second Prizes drawn.

Questions:

(1) Construct the joint probability distribution table for (X,Y) and verify its validity.

(2) Find the marginal distributions of X and Y, and determine if they are independent.

(3) Define $Z = X + Y$. Find the probability distribution of Z.

4. 设$X \sim N(0,1)$.

(1) 求$Y = e^X$ 的概率密度函数.

(2) 求$Y = X^2 + 1$ 的概率密度函数.

Let $X \sim N(0,1)$.

(1) Find the probability density function of $Y = e^X$.

(2) Find the probability density function of $Y = X^2 + 1$.

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5. 假设随机变量X和Y 独立, 且分别服从泊松分布$X \sim P(\lambda_1)$, $Y \sim P(\lambda_2)$.

(1) 求X 和Y 的联合频率函数.

(2) 求条件概率$P(X = k \mid X + Y = n)$, 其中$n \geq k$ 是非负整数.

Suppose random variables X and Y are independent and follow Poisson distributions $X \sim P(\lambda_1)$ and $Y \sim P(\lambda_2)$, respectively.

(1) Find the joint probability mass function of X and Y;

(2) Find the conditional probability $P(X = k \mid X + Y = n)$, where $n \geq k$ is a non-negative integer.

$P(A|B) = P(A)/P(B)$

6. 令随机向量(X, Y) 的联合分布函数为

$$F(x,y) = \begin{cases} c(1-e^{-2x})(1-e^{-3y}), & 0 < x, 0 < y, \\ 0, & \text{otherwise}, \end{cases}$$

$e^{-2x}\Big|_1^3$

$e^{-2} - e^{-6}$

其中c 为常数.

(1) 求出常数c 及X 和Y 的联合概率密度函数f.

(2) 求出X 与Y 的边缘概率密度函数f_X 和f_Y. 试判断X 和Y 是否独立, 并说明理由.

(3) 计算$P(1 < X < 3, 1 < Y < 2)$.

Let (X, Y) be a random vector with joint c.d.f.

$$F(x,y) = \begin{cases} c(1-e^{-2x})(1-e^{-3y}), & 0 < x, 0 < y, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant.

(1) Determine the value of c and find the joint p.d.f. f of X and Y.

(2) Find the two marginal p.d.f.s f_X and f_Y, and determine with reasons whether X and Y are independent.

(3) Calculate $P(1 < X < 3, 1 < Y < 2)$.

$\frac{n!}{k_1!(n-k_1)!} \cdot \lambda_1^{k_1}$

$(\lambda_1+\lambda_2)^n$.

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